

Math 421: Homework 9

Solutions

Due: Nov 21, 2025

Problem 1: (Ross 17.1) - Solutions given in back of textbook.

Problem 2: Prove, using the $\epsilon - \delta$ definition of continuity, the following theorem.

Theorem. *Let f and g be real-valued functions that are continuous at $x_0 \in \mathbb{R}$. Then $f + g$ is continuous at x_0 .*

Proof. Since $x_0 \in \text{dom}(f) \cap \text{dom}(g)$, we have $x_0 \in \text{dom}(f + g)$. Let $\epsilon > 0$. Since f is continuous at x_0 , there exists $\delta_1 > 0$ such that if $|x - x_0| < \delta_1$ for all $x \in \text{dom}(f)$, then $|f(x) - f(x_0)| < \epsilon/2$. Since g is continuous at x_0 , there exists $\delta_2 > 0$ such that if $|x - x_0| < \delta_2$ for all $x \in \text{dom}(g)$ then $|g(x) - g(x_0)| < \epsilon/2$. Let $\delta = \min\{\delta_1, \delta_2\}$. Assume $x \in \text{dom}(f + g)$ and $|x - x_0| < \delta$. Then

$$\begin{aligned} |(f + g)(x) - (f + g)(x_0)| &= |f(x) + g(x) - (f(x_0) + g(x_0))| \\ &\leq |f(x) - f(x_0)| + |g(x) - g(x_0)| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon. \end{aligned}$$

Since $\epsilon > 0$ was arbitrary, $f + g$ is continuous at x_0 . □

Problem 3: Prove, using the limit theorems for sequences, the following theorem.

Theorem. *Let f and g be real-valued functions that are continuous at $x_0 \in \mathbb{R}$. Then fg is continuous at x_0 .*

Proof. Since $x_0 \in \text{dom}(f) \cap \text{dom}(g)$, we have $x_0 \in \text{dom}(fg)$. Let (x_n) be a sequence such that $x_n \in \text{dom}(fg)$ for all n and $\lim x_n = x_0$. Then, since f and g are both continuous at x_0 , we have $\lim f(x_n) = f(x_0)$ and $\lim g(x_n) = g(x_0)$. So

$$\lim(fg)(x_n) = \lim[f(x_n)g(x_n)] = (\lim f(x_n))(\lim g(x_n)) = f(x_0)g(x_0) = (fg)(x_0)$$

by the algebraic limit laws for sequences. Thus, fg is continuous at x_0 . □

Problem 4: (Ross 17.8) Let f and g be real-valued functions.

(a) Show $\min(f, g) = \frac{1}{2}(f + g) - \frac{1}{2}|f - g|$.

Proof. First we prove that if $a, b \in \mathbb{R}$, then $\min(a, b) = \frac{1}{2}(a + b) - \frac{1}{2}|a - b|$.

Case 1: Assume $a \leq b$. Then $|a - b| = -(a - b)$, so that

$$\frac{1}{2}(a + b) - \frac{1}{2}|a - b| = \frac{1}{2}(a + b) + \frac{1}{2}(a - b) = a = \min(a, b).$$

Case 2: Assume $b < a$. Then $|a - b| = a - b$, so that

$$\frac{1}{2}(a + b) - \frac{1}{2}|a - b| = \frac{1}{2}(a + b) - \frac{1}{2}(a - b) = b = \min(a, b).$$

Then we have $\min(f(x), g(x)) = \frac{1}{2}(f(x) + g(x)) - \frac{1}{2}|f(x) - g(x)|$ and so $\min(f, g) = \frac{1}{2}(f + g) - \frac{1}{2}|f - g|$. □

(b) Show $\min(f, g) = -\max(-f, -g)$.

Proof. From Example 5, we have

$$\begin{aligned}
 -\max(-f, -g) &= -\left(\frac{1}{2}(-f + (-g)) + \frac{1}{2}|-f - (-g)|\right) \\
 &= -\frac{1}{2}(-f - g) - \frac{1}{2}|-f + g| \\
 &= \frac{1}{2}(f + g) - \frac{1}{2}|-(f - g)| \\
 &= \frac{1}{2}(f + g) - \frac{1}{2}|f - g| \\
 &= \min(f, g)
 \end{aligned}$$

□

(c) Let f and g be continuous at $x_0 \in \mathbb{R}$. Prove $\min(f, g)$ is continuous at x_0 .

Proof. Note that, from part (a), $\min(a, b) = \frac{1}{2}(a + b) - \frac{1}{2}|a - b|$. From Theorem 17.4 (i), $f + g$ and $f - g$ are continuous at x_0 . Hence $|f - g|$ at x_0 by Theorem 17.3, as well as $\frac{1}{2}(f + g)$ and $\frac{1}{2}|f - g|$. Thus, applying Theorem 17.4 again, we have $\frac{1}{2}(a + b) - \frac{1}{2}|a - b| = \min(a, b)$ is continuous at x_0 . □

Problem 5: (Ross 17.12)

(a) Let f be a continuous real-valued function with domain (a, b) . Show that if $f(r) = 0$ for each rational number r in (a, b) , then $f(x) = 0$ for all $x \in (a, b)$.

Proof. Let f be a continuous real-valued function with domain (a, b) . Let $x \in (a, b)$. There exists a sequence (r_n) of rational numbers such that $\lim r_n = x$. By the definition of continuity, we have

$$\lim f(r_n) = \lim 0 = 0 = f(x)$$

for all $x \in (a, b)$.

□

(b) Let f and g be continuous real-valued functions on (a, b) such that $f(r) = g(r)$ for each rational number r in (a, b) . Prove that $f(x) = g(x)$ for all $x \in (a, b)$.

Proof. Let f and g be continuous real-valued functions on (a, b) such that $f(r) = g(r)$. Define $h = f - g$, h is then also a continuous function on (a, b) and $h(r) = f(r) - g(r) = 0$ for each rational number r in (a, b) . Then, by part (a), we have $h(x) = f(x) - g(x) = 0$ for all $x \in (a, b)$, so that $f(x) = g(x)$. □

Problem 6: (Ross 17.13b) Let

$$h(x) = \begin{cases} x & x \in \mathbb{Q} \\ 0 & x \notin \mathbb{Q} \end{cases}$$

show h is continuous at $x = 0$ and at no other point.

Proof. Let $x_0 = 0$, let (s_n) be a sequence such that $\lim s_n = 0$. For all $\epsilon > 0$ there exists N such that for $n > N$ we have $|s_n - 0| < \epsilon$. Consider $(h(s_n))$, if s_n is rational, we have $|h(s_n) - 0| = |s_n - 0| < \epsilon$, and if s_n is irrational, we have $|h(s_n) - 0| = |0 - 0| < \epsilon$. Therefore, $\lim h(s_n) = 0 = h(0)$.

Let $x_0 \neq 0$. If $x_0 \in \mathbb{Q}$, then let (s_n) be the sequence defined by $s_n = x_0 - \frac{\sqrt{2}}{n}$, a sequence of irrational numbers converging to x_0 . Then $\lim h(s_n) = \lim 0 = 0 \neq x_0 = f(x_0)$. If $x_0 \notin \mathbb{Q}$, and let (s_n) be a sequence of rational numbers converging to x_0 , then $\lim h(s_n) = \lim s_n = x_0 \neq 0 = h(x_0)$. □